Astrology: Science or Pseudoscience?

This is an in-class lab activity for `ASTR170B1 - Introduction to Astronomy`, to be completed in groups of 2-3 people, but you should each turn in your own results. You should answer all of the questions in the space provided. Each group member should participate equally in the lab. If you finish all of the questions, you may turn in your lab in class today. Otherwise, this assignment is due *<2026-09-09 Wed>* at the start of class.

Your name ____________________ **and student ID** ____________________

Aim: *Introduce the concept of pseudoscience and to identify the qualities that distinguish it from legitimate science.*

Case study: Astrology

Initial question

What is science? What is pseudoscience? Why is it important to know the distinction as you navigate today's world? *Discuss with your classmates and briefly note your initial thoughts on this topic.* How would you explain the difference between science and pseudoscience to a friend? (max 4 lines)

Astrology

There are various astrological traditions throughout the world. Most originated in a context where it was intrinsically intertwined with astronomy. Both disciplines require to measure the positions of celestial bodies in the sky as a function of time - a need still very much existing today, with some of the most advanced space telescopes (e.g., the *Gaia* satellite) pushing technological limits of today to achieve unprecedented precision in this very same task.

However, while astronomy was always concerned with the explanation of the motion observed in the sky, and its prediction, all astrology is based on *assuming* that the positions of celestial objects at the time of one's birth (or other special times in ones life) influence their person and behavior.

Note how this is not an absurd hypothesis to begin with: the position of the Sun and Moon do influence phenomena on Earth (is it day or not? Is it high-tide or low-tide?), and the basic hypothesis of astrology just extends this to all celestial bodies.

Applying the scientific method

Step 1: Make an observation

This class contains students with different personalities and life experiences, born at different times and in different locations. As we learned in class, the stars, planets, and moon appear to move in the sky, so their positions at each of our births was different. These are both observational facts; to test astrology, we need to design an experiment to test if one can explain the other.

Step 2: Suggest an hypothesis

An hypothesis here is just a "guess", typically educated by your existing knowledge. In this case we can make two opposite hypothesis: *The positions of the sun, moon, planets, and other astronomical objects at the time of one's birth*

☐ does (= astrology works)

☐ does **not** (=astrology does **not** work)

affect one's personality and predict one's future.

Select one of the two options above. Note that a priori neither is right or wrong. We will have to make our own test of it.

Step 3: Based on the hypothesis, make a prediction

I have collected here horoscopes from 3 sources for yesterday *<2026-09-01 Tue>* and removed the "signs", i.e. the birthdate windows that indicate which horoscope is yours. Each of you will read through all of the horoscopes and choose the 1 from each set that best describes your day.

For reference, these are the signs (recall the connection with zodiacal constellations):

Sign	Birthdates
Aries	March 21 – April 19
Taurus	April 20 – May 20
Gemini	May 21 – June 20
Cancer	June 21 – July 22
Leo	July 23 – August 22
Virgo	August 23 – September 22
Libra	September 23 – October 22
Scorpio	October 23 – November 21
Sagittarius	November 22 – December 19
Capricorn	December 22 – January 19
Aquarius	January 20 – February 18
Pisces	February 19 – March 20

Circle the sign corresponding to your birthday, and based on the hypothesis you selected above (does/does not), what do you predict will be the outcome of this experiment? (max 3 lines)

Step 4: Perform a test (the experiment)

First read through the three sets of horoscopes for <*2026-09-02 Wed*>. Write down the number for each horoscope that best applies to you:

Set 1: ______ Set 2: ______ Set 3: ______

Now, I will reveal the sign that corresponds to each horoscope. Write down the signs that correspond to the numbers you selected above:

Sign 1: ______ Sign 2: ______ Sign 3: ______

How many of the horoscopes did you identify correctly? ______

Now, we will collect data from the entire class. Please add your result to the Top Hat poll and copy results for the full class below. Percentage of students who correctly identified:

0 horoscopes: ______ 1 horoscope: ______ 2 horoscopes: ______ 3 horoscopes: ______

Step 5: Analyze the results

If astrology is correct, we would expect everyone to pick all of the answers that correspond to their sign. But if astrology is incorrect, does that mean everyone will choose all of the wrong horoscopes? Of course not – it is possible to pick the correct horoscope by chance.

We can compute the probability that you chose correctly by pure chance any number of times using the Binomial Distribution: the probability (P) for k successes from N trials where the probability of success is p is given by:

$$P(k, N, p) = \frac{N!}{k!(N-k)!} p^k (1-p)^{n-k} \quad , \tag{1}$$

where that exclamation point ! indicates "factorial", that is is the product of all integers equal to or less than a number. For instance, $4! = 4 \times 3 \times 2 \times 1 = 24$, and by definition, $0! = 1$.

The probability of success p is the chance that you achieve an outcome once in one trial. For example, if you flip a fair coin, the probability of it coming up heads is one in two, or $p = 1/2 = 0.5$. So the probability of having say $k = 7$ heads out of $N = 10$ launches is: $P(k = 7, N = 10, p = 0.5) = \frac{10!}{7!3!}(0.5)^{0.7}(0.5)^{0.3} \simeq 0.11$.

- What are p and N in our experiment? ____________

- What is the probability that you would choose zero, one, two or three correct horoscopes by guessing at random? ____________

Hint: use the binomial theorem with k = 0, 1, 2, and 3. This online calculator may also be helpful: `https://www.desmos.com/scientific`

- Comparing with your classmates, do your correct guess numbers follow the binomial distribution? ____________

- What do you conclude about the efficacy of horoscopes? Does this outcome support your original hypothesis? If not, create a new one.

Additional questions

Science

Science seeks explanations that rely solely on natural causes and progresses through the creation and testing of models of nature. A theory, model, prediction, measurement, observation, etc. is “scientific” if it has these characteristics:

- Repeatable
- Objective
- Predictive
- Falsifiable (can be tested and updated if necessary!)

A scientific theory is a simple yet powerful model that explains a wide variety of observations using just a few general principles and has been verified by repeated and varied testing.” Theories begin as hypotheses: testable predictions that can be corroborated or falsified via the scientific method. One common misconception is that scientific theories are "just theories".

Can you think of a modern instance in which people have incorrectly argued that a scientific theory is “just a theory” with no more validity than an alternative belief? (max 4 lines)

Pseudoscience

The term ‘pseudoscience’ refers to any belief, claim, practice, etc. that is presented as being ‘scientific’, but does not follow the scientific method or possess the characteristics discussed above. There are many examples of pseudoscience in our modern society. Pseudoscientific ideas range from the relatively harmless (such as cryptozoology, the belief that creatures like the Loch Ness monster and unicorns exist) to the downright evil (such as “scientific racism”, the belief that certain races of humans are biologically superior).

- How easy is it to compare the different horoscope “predictions” to your lived experience (and why)? Would you classify astrology as science or pseudoscience? (max 4 lines)

- Think of two more examples of pseudoscience and explain why they are pseudoscientific. (max 4 lines)

Summary table

Science
based on objective quantitative observations
makes quantitative and testable predictions
Prefers simple explanations exclusively based on natural phenomena
Pseudoscience
not necessarily based on *objective* or *quantitative* observations
makes vague predictions adaptable to many outcomes
does not change if new evidence emerges